

The leading coefficient of the graded 2-core law:
a refined 2-quotient character identity and the closed form

$$S(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n)$$

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Abstract

For $\lambda \vdash n$ let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$ be the graded branch-exponent enumerator, $s(T) = \sum_{i \in \text{Des}(T)} w_i$ with parity-twisted weights $w_i = (2i - 1)[n - i \text{ odd}]$. The graded 2-core law ($\text{ord}_{q=-1} G_\lambda = c := \lfloor c_0/2 \rfloor$, $c_0 = |2\text{-core}(\lambda)|$) is proved, and its leading Taylor coefficient at $q = -1$ was reduced to one λ -dependent character value $\chi^\lambda(2^w 1^{c_0})$, $w = (n - c_0)/2$. Here I close that last piece. I prove the *refined 2-quotient character identity*

$$\chi^\lambda(2^w 1^{c_0}) = \varepsilon(c_0) (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{2\text{-core}(\lambda)},$$

where $(\lambda^{(0)}, \lambda^{(1)})$ is the 2-quotient and the sign $\varepsilon(c_0) = (-1)^{n(\delta_k)} = (-1)^{\binom{k+1}{3}}$ depends only on the 2-core $\delta_k = (k, k - 1, \dots, 1)$ (so on $c_0 = \binom{k+1}{2}$), being -1 exactly when $k \equiv 2 \pmod{4}$. The proof is one iterated application of the Murnaghan–Nakayama rule by dominoes: domino removal preserves the 2-core, so the strip-tableau sum collapses onto the single core term, whose signed count factors through the Stanton–White correspondence with a sign that is constant by a row-parity invariant. Assembling with the master identity yields the leading coefficient in closed form,

$$a_c = (-1)^{n(\lambda)} \varepsilon(c_0) 2^{-c} e_c(W_n) f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{2\text{-core}(\lambda)}, \quad S(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n),$$

with W_n the arithmetic progression $\{4j + a : 0 \leq j < N\}$ ($(a, N) = (1, n/2)$ for n even, $(3, (n - 1)/2)$ for n odd); I give $e_c(W_n)$ in closed (Pochhammer / generalized-Stirling) form. The single-valuedness of $S(n, c)$ across all $\lambda \vdash n$ with fixed (n, c) is exactly the content of the character identity. All formulas are verified by exact computation for every partition of every $n \leq 12$ (270 shapes, 0 mismatches).

1 Statement and setup

I keep the notation of the graded 2-core law (“06-01”). Throughout $\lambda \vdash n$; $\text{SYT}(\lambda)$, $\text{Des}(T)$, $\text{comaj}(T) = \sum_{i \in \text{Des}(T)} (n - i)$, $n(\lambda) = \sum_i (i - 1)\lambda_i$, $f^\mu = \#\text{SYT}(\mu)$ are standard. The branch weights are $w_i = (2i - 1)[n - i \text{ odd}]$, the *active set* is $A = \{k \in [n - 1] : n - k \text{ odd}\}$ with $|A| = m = \lfloor n/2 \rfloor$, and

$$G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}, \quad s(T) = \sum_{i \in \text{Des}(T)} w_i.$$

Every λ has a 2-core, which is a staircase $\delta_k = (k, k-1, \dots, 1)$ of size $c_0 := |2\text{-core}(\lambda)| = \binom{k+1}{2}$, and a 2-quotient $(\lambda^{(0)}, \lambda^{(1)})$ with $|\lambda^{(0)}| + |\lambda^{(1)}| = w := (n - c_0)/2$ (the number of dominoes). Set $c = \lfloor c_0/2 \rfloor$, so $c_0 = 2c + e$ with $e = n \bmod 2$, and note $m - c = w$.

What is already proved (06-01). The order law $\text{ord}_{q=-1} G_\lambda = c$ holds, and the first nonzero Taylor coefficient a_c in the expansion $G_\lambda(q) = \sum_k a_k(q+1)^k$ is

$$a_c = (-1)^c e_c(W_n) \left(-\frac{1}{2}\right)^c \chi^\lambda(2^w 1^{c_0}) = 2^{-c} e_c(W_n) \chi^\lambda(2^w 1^{c_0}), \quad (1)$$

where

$$W_n = \{2k - 1 : k \in A\} \quad (2)$$

and e_c is the elementary symmetric polynomial. (The two forms agree because $(-1)^c (-\frac{1}{2})^c = 2^{-c}$.) Equation (1) is line-for-line the output of the 06-01 master identity; the only object in it that still depended on λ in an unresolved way is the character value $\chi^\lambda(2^w 1^{c_0})$ — a class with w transpositions and c_0 fixed points.

Goal. Evaluate $\chi^\lambda(2^w 1^{c_0})$ in closed form (Theorem 1), deduce the leading coefficient and $S(n, c)$ (Theorem 2), pin the sign $\varepsilon(c_0)$ (Proposition 1), and give $e_c(W_n)$ in closed form (Proposition 2).

2 The refined 2-quotient character identity

Theorem 1. Let $\lambda \vdash n$ have 2-core δ_k of size $c_0 = \binom{k+1}{2}$, 2-quotient $(\lambda^{(0)}, \lambda^{(1)})$, and $w = (n - c_0)/2$ dominoes. Then

$$\chi^\lambda(2^w 1^{c_0}) = \varepsilon(c_0) (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k}, \quad \varepsilon(c_0) = (-1)^{n(\delta_k)} \in \{\pm 1\}.$$

In particular the value is nonzero, and depends on λ only through $(-1)^{n(\lambda)}$ and the 2-quotient data; the sign $\varepsilon(c_0)$ depends only on c_0 .

This generalizes the $c_0 \in \{0, 1\}$ case (the reversal/longest-element class w_0), proved in “06-03”, where $\delta_k \in \{\emptyset, (1)\}$, $f^{\delta_k} = 1$, and $\varepsilon = +1$. The proof is the same engine — iterated domino Murnaghan–Nakayama — carried out over a nonempty staircase core.

2.1 Collapse onto the core: only $\mu = \delta_k$ survives

Lemma 1. Apply the Murnaghan–Nakayama (MN) rule to χ^λ on the class $2^w 1^{c_0}$, removing the w parts of size 2 first and the c_0 fixed points last. Then

$$\chi^\lambda(2^w 1^{c_0}) = N(\lambda) \cdot f^{\delta_k}, \quad N(\lambda) = \sum_{\substack{\lambda = \nu_0 \supset \nu_1 \supset \dots \supset \nu_w = \delta_k \\ \nu_{i-1}/\nu_i \text{ a domino}}} \prod_{i=1}^w (-1)^{\text{ht}(\nu_{i-1}/\nu_i) - 1},$$

the signed count of domino-strip removal sequences from λ down to its 2-core, where ht is the number of rows a strip occupies.

Proof. The MN rule, with the cycle parts ordered as w twos then c_0 ones, gives

$$\chi^\lambda(2^w 1^{c_0}) = \sum_{\lambda = \nu_0 \supset \dots \supset \nu_w} \left(\prod_{i=1}^w (-1)^{\text{ht}(\nu_{i-1}/\nu_i) - 1} \right) \chi^{\nu_w}(1^{c_0}),$$

the sum over chains in which each skew ν_{i-1}/ν_i is a border strip (rim hook) of size 2, i.e. a domino, so $|\nu_w| = n - 2w = c_0$; and $\chi^{\nu_w}(1^{c_0}) = f^{\nu_w}$ (removing c_0 single cells, all of height 1, contributes only $+1$'s, and counts SYT). It remains to see that the only $\nu_w \vdash c_0$ that occurs is δ_k .

Removing a domino (a 2-rim-hook) does not change the 2-core of a partition (James–Kerber, *The Representation Theory of the Symmetric Group*, §2.7). Hence every ν_i in such a chain has the same 2-core as λ , namely δ_k ; in particular $2\text{-core}(\nu_w) = \delta_k$, so $|2\text{-core}(\nu_w)| = c_0 = |\nu_w|$. A partition whose 2-core has its own size is equal to its 2-core (the core is reached by removing dominoes, each lowering the size by 2; equal size forces zero removals). Therefore $\nu_w = \delta_k$, and $\chi^{\nu_w}(1^{c_0}) = f^{\delta_k}$ in every surviving term. Factoring it out gives the claim. $\square$

A removal chain $\lambda = \nu_0 \supset \cdots \supset \nu_w = \delta_k$ is the same datum as a *standard domino tableau* of the skew shape λ/δ_k : tile λ/δ_k by w dominoes and number them $1, \dots, w$ by reverse removal order. The MN weight of a chain is

$$\prod_{i=1}^w (-1)^{\text{ht}(\nu_{i-1}/\nu_i)-1} = (-1)^{v(D)},$$

where $v(D)$ is the number of *vertical* dominoes of the underlying tiling D (a horizontal domino has $\text{ht} = 1$, a vertical one has $\text{ht} = 2$). This weight depends only on the tiling D , not on the numbering. So

$$N(\lambda) = \sum_D (-1)^{v(D)} \#\{\text{standard fillings of } D\}, \quad (3)$$

the sum over domino tilings D of λ/δ_k .

2.2 The sign is constant: a row-parity invariant

Lemma 2. *For every domino tiling D of the skew shape λ/δ_k ,*

$$v(D) \equiv n(\lambda) + n(\delta_k) \pmod{2}.$$

In particular $(-1)^{v(D)}$ is the same for all tilings, equal to $(-1)^{n(\lambda)+n(\delta_k)}$.

Proof. Colour a cell (i, j) by the parity of its row index i . A horizontal domino occupies $(i, j), (i, j+1)$ — two cells of the *same* row-parity; a vertical domino occupies $(i, j), (i+1, j)$ — one cell of each row-parity. Let a be the number of cells of λ/δ_k lying in odd rows. Counting odd-row cells by domino type,

$$a = v(D) \cdot 1 + h_{\text{odd}}(D) \cdot 2,$$

where $h_{\text{odd}}(D)$ is the number of horizontal dominoes in odd rows (each vertical domino contributes exactly one odd-row cell; a horizontal domino contributes 2 if it sits in an odd row and 0 otherwise). Hence $v(D) \equiv a \pmod{2}$, and the right-hand side a depends only on λ/δ_k , not on D .

It remains to evaluate $a \pmod{2}$. For any partition μ , writing $b_\mu = \sum_{i \text{ even}} \mu_i$ for the number of cells of μ in even rows, we have

$$n(\mu) = \sum_i (i-1)\mu_i \equiv \sum_{i \text{ even}} \mu_i = b_\mu \pmod{2},$$

since $i-1$ is odd exactly when i is even. The number of odd-row cells of μ is $|\mu| - b_\mu$. Therefore $a = (n - b_\lambda) - (c_0 - b_{\delta_k}) = (n - c_0) - b_\lambda + b_{\delta_k} = 2w - b_\lambda + b_{\delta_k} \equiv b_\lambda + b_{\delta_k} \equiv n(\lambda) + n(\delta_k) \pmod{2}$, which gives the claim. $\square$

Remark 1. Lemma 2 is the skew-shape form of the spin/2-quotient parity invariant; over an empty core it recovers the “06-03” statement $v(D) \equiv n(\lambda) \pmod{2}$. The row-parity count makes both the constancy and the value visible at once, with no appeal to the Stanton–White bijection.

2.3 The count: Stanton–White

Lemma 3. *The number of standard domino tableaux of λ (equivalently, of standard fillings of all domino tilings of λ/δ_k) is*

$$\# \text{SDT}(\lambda) = f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|}.$$

Proof. This is the Stanton–White correspondence (James–Kerber §2.7; Stanton–White, *J. Combin. Theory A* 40 (1985)). It is a bijection between standard domino tableaux of λ and triples $(P^{(0)}, P^{(1)}, \sigma)$, where $P^{(r)} \in \text{SYT}(\lambda^{(r)})$ and σ is a shuffle distributing the w time labels $\{1, \dots, w\}$ among the two quotient components, $|\lambda^{(0)}|$ to the first and $|\lambda^{(1)}|$ to the second. The number of such shuffles is $\binom{w}{|\lambda^{(0)}|}$, giving the product. (The bijection is purely combinatorial and holds for every λ , irrespective of its core.) $\square$

2.4 Assembly of Theorem 1

By (3) and Lemma 2, the sign is constant over tilings, so

$$N(\lambda) = (-1)^{n(\lambda)+n(\delta_k)} \sum_D \#\{\text{standard fillings of } D\} = (-1)^{n(\lambda)+n(\delta_k)} \# \text{SDT}(\lambda).$$

Combining with Lemma 1 and Lemma 3,

$$\chi^\lambda(2^w 1^{c_0}) = N(\lambda) f^{\delta_k} = (-1)^{n(\delta_k)} (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k}.$$

Setting $\varepsilon(c_0) = (-1)^{n(\delta_k)}$ — which depends only on k , hence only on $c_0 = \binom{k+1}{2}$ — proves Theorem 1. The value is nonzero because $f^{\lambda^{(0)}}, f^{\lambda^{(1)}}, f^{\delta_k} \geq 1$ and $\binom{w}{|\lambda^{(0)}|} \geq 1$. $\square$

3 The leading coefficient and $S(n, c)$

Theorem 2. *With W_n as in (2), the leading Taylor coefficient of G_λ at $q = -1$ is*

$$a_c = (-1)^{n(\lambda)} \varepsilon(c_0) 2^{-c} e_c(W_n) f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k}.$$

Equivalently, writing $a_c = (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k} S(n, c)$,

$$\boxed{S(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n)}.$$

The scalar $S(n, c)$ depends only on (n, c) ; this single-valuedness over all $\lambda \vdash n$ with 2-core size $c_0 = 2c + e$ is precisely Theorem 1.

Proof. Substitute Theorem 1 into the master form (1):

$$a_c = 2^{-c} e_c(W_n) \varepsilon(c_0) (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k}.$$

Dividing by $(-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} f^{\delta_k}$ (each factor nonzero) isolates $S(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n)$, manifestly a function of (n, c) alone. $\square$

Remark 2 (Two conventions for S). Some notes fold f^{δ_k} into S , writing $a_c = (-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \binom{w}{|\lambda^{(0)}|} \tilde{S}(n, c)$ with $\tilde{S}(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n) f^{\delta_k}$; then $\tilde{S}(n, 1) = -\binom{n}{2}$ (since $f^{\delta_2} = f^{(2,1)} = 2$). The convention of Theorem 2 keeps f^{δ_k} separate, giving the cleaner $S(n, 1) = -\frac{1}{2}\binom{n}{2}$. Both are equivalent statements about the same a_c .

4 The sign $\varepsilon(c_0)$ in closed form

Proposition 1. *For the staircase core $\delta_k = (k, k-1, \dots, 1)$ with $c_0 = \binom{k+1}{2}$,*

$$\varepsilon(c_0) = (-1)^{n(\delta_k)} = (-1)^{\binom{k+1}{3}},$$

and $\varepsilon(c_0) = -1$ if and only if $k \equiv 2 \pmod{4}$.

Proof. Compute $n(\delta_k)$. With $\delta_{k,i} = k - i + 1$,

$$n(\delta_k) = \sum_{i=1}^k (i-1)(k-i+1) = \sum_{j=0}^{k-1} j(k-j) = k \frac{(k-1)k}{2} - \frac{(k-1)k(2k-1)}{6} = \frac{(k-1)k(k+1)}{6} = \binom{k+1}{3}.$$

Hence $\varepsilon(c_0) = (-1)^{\binom{k+1}{3}}$. For the parity criterion, by Lucas' theorem $\binom{k+1}{3}$ is odd iff the binary digits of $3 = (11)_2$ lie under those of $k+1$, i.e. iff the last two bits of $k+1$ are both 1, i.e. $k+1 \equiv 3 \pmod{4}$, i.e. $k \equiv 2 \pmod{4}$. $\square$

Thus $\varepsilon = +1$ for $k \in \{0, 1, 3, 4, 5, 7, 8, 9, \dots\}$ and $\varepsilon = -1$ for $k \in \{2, 6, 10, \dots\}$; in c -terms this is $\varepsilon(c_0) = -1$ exactly at $c \in \{1, 10, \dots\}$ (i.e. $c_0 \in \{3, 21, \dots\}$), matching the observed $\varepsilon = -1$ at $c = 1$ and $\varepsilon = +1$ at $c \in \{0, 3, 5, 7\}$.

5 Closed form for $e_c(W_n)$

Proposition 2. *The set $W_n = \{2k-1 : k \in A\}$ is the arithmetic progression*

$$W_n = \{4j + a : 0 \leq j < N\}, \quad (a, N) = \begin{cases} (1, n/2), & n \text{ even}, \\ (3, (n-1)/2), & n \text{ odd}. \end{cases}$$

Consequently its elementary symmetric polynomials have generating function

$$\sum_{c \geq 0} e_c(W_n) t^c = \prod_{j=0}^{N-1} (1 + (4j + a)t) = (4t)^N \left(\frac{a}{4} + \frac{1}{4t} \right)_N,$$

where $(z)_N = z(z+1) \cdots (z+N-1)$ is the Pochhammer (rising-factorial) symbol; equivalently, via the unsigned Stirling numbers of the first kind $\left[\begin{smallmatrix} N \\ \kappa \end{smallmatrix} \right]$,

$$e_c(W_n) = \sum_{\kappa=0}^N \left[\begin{smallmatrix} N \\ \kappa \end{smallmatrix} \right] 4^{N-\kappa} a^{c-N+\kappa} \binom{\kappa}{c-N+\kappa},$$

the sum running over κ with $0 \leq c - N + \kappa \leq \kappa$.

Proof. If n is even then $n-k$ is odd iff k is odd, so $A = \{1, 3, \dots, n-1\}$ and $W_n = \{1, 5, \dots, 2n-3\} = \{4j+1 : 0 \leq j < n/2\}$. If n is odd then $n-k$ is odd iff k is even, so $A = \{2, 4, \dots, n-1\}$ and $W_n = \{3, 7, \dots, 2n-3\} = \{4j+3 : 0 \leq j < (n-1)/2\}$. This is the displayed progression with common difference 4.

The first factorization is the definition of e_c as the coefficients of $\prod_{w \in W_n} (1 + wt)$. For the Pochhammer form, write $1 + (4j + a)t = 4t(j + \frac{1+at}{4t})$ and use $\prod_{j=0}^{N-1} (j + z) = (z)_N$ with $z = \frac{1+at}{4t} = \frac{a}{4} + \frac{1}{4t}$. For the Stirling form, expand $(z)_N = \sum_{\kappa} \left[\begin{smallmatrix} N \\ \kappa \end{smallmatrix} \right] z^\kappa$ and substitute $z = \frac{a}{4} + \frac{1}{4t}$, i.e.

$$\prod_{j=0}^{N-1} (1 + (4j + a)t) = \sum_{\kappa=0}^N \left[\begin{smallmatrix} N \\ \kappa \end{smallmatrix} \right] (4t)^{N-\kappa} (1 + at)^\kappa;$$

reading off $[t^c]$ (the term needs $t^{N-\kappa} \cdot t^{c-N+\kappa}$ from $(1+at)^\kappa$) gives the stated convolution. $\square$

For instance $e_1(W_n) = \sum_{w \in W_n} w = \binom{n}{2}$ (whence $S(n, 1) = -\frac{1}{2}\binom{n}{2}$), and the smallest higher values are $e_3(W_6) = 1 \cdot 5 \cdot 9 = 45$ ($S(6, 3) = 45/8$), $e_5(W_{10}) = 1 \cdot 5 \cdot 9 \cdot 13 \cdot 17 = 9945$ ($S(10, 5) = 9945/32$).

6 Computational verification

All claims were checked by independent exact (rational) computation; the SYT side (direct enumeration of $\text{SYT}(\lambda)$ with $s(T)$, Taylor expansion of G_λ about $q = -1$) shares no code with the character/2-quotient side.

- **Leading coefficient (Theorem 2).** For all $\lambda \vdash n$, $2 \leq n \leq 12$ (270 shapes), the directly computed a_c equals $(-1)^{n(\lambda)} \varepsilon(c_0) 2^{-c} e_c(W_n) f^{\lambda^{(0)}} f^{\lambda^{(1)}} \left(\begin{smallmatrix} w \\ |\lambda^{(0)}| \end{smallmatrix} \right) f^{\delta_k}$: 0 mismatches.
- **Single-valuedness / character identity (Theorem 1).** The reduced scalar $a_c / ((-1)^{n(\lambda)} f^{\lambda^{(0)}} f^{\lambda^{(1)}} \left(\begin{smallmatrix} w \\ |\lambda^{(0)}| \end{smallmatrix} \right) f^{\delta_k})$ is constant across all $\lambda \vdash n$ of each fixed (n, c) , $n \leq 12$, and equals $\varepsilon(c_0) 2^{-c} e_c(W_n)$. Sample table $(S(n, c))$: $S(n, 1) = -\frac{1}{2}\binom{n}{2}$; $S(6, 3) = 45/8$, $S(8, 3) = 203/2$, $S(10, 3) = 2565/4$, $S(12, 3) = 5115/2$; $S(10, 5) = 9945/32$, $S(12, 5) = 156057/16$.
- **Sign (Proposition 1).** $\varepsilon(c_0) = (-1)^{n(\delta_k)} = (-1)^{\binom{k+1}{3}}$ for $k = 0, \dots, 6$: $\varepsilon = +1$ except $k \in \{2, 6\}$ where $\varepsilon = -1$ (matching $c \in \{1, 10\}$).
- **e_c closed forms (Proposition 2).** The AP description, the Pochhammer generating function, and the Stirling convolution all reproduce $e_c(W_n)$ exactly for every $n \leq 12$ and every c .

Scripts. `~/projects/scratch/2026-06-05-Snc/` (notably `snc12.py` for the single-valued $S(n, c)$ table, `verify_closedform.py` for the full a_c identity and the sign, `verify_ec.py` for the e_c closed forms), reusing the SYT/2-quotient/Murnaghan–Nakayama routines of `~/projects/scratch/prove-2026-06-01-`

7 Summary

The graded 2-core law $\text{ord}_{q=-1} G_\lambda = c$ is now complete down to its leading coefficient. The one remaining λ -dependent quantity, $\chi^\lambda(2^w 1^{c_0})$, is evaluated in closed form (Theorem 1) by iterated domino Murnaghan–Nakayama: domino removal preserves the 2-core, collapsing the strip-tableau sum onto the single core term δ_k ; the resulting signed domino-tableau count factors through Stanton–White, with a sign made constant by a one-line row-parity invariant. Assembling with the master identity gives

$$a_c = (-1)^{n(\lambda)} \varepsilon(c_0) 2^{-c} e_c(W_n) f^{\lambda^{(0)}} f^{\lambda^{(1)}} \left(\begin{smallmatrix} w \\ |\lambda^{(0)}| \end{smallmatrix} \right) f^{\delta_k}, \quad S(n, c) = \varepsilon(c_0) 2^{-c} e_c(W_n),$$

with $\varepsilon(c_0) = (-1)^{\binom{k+1}{3}}$ (-1 iff $k \equiv 2 \pmod{4}$, $c_0 = \binom{k+1}{2}$) and $e_c(W_n)$ the elementary symmetric polynomial of the arithmetic progression $\{4j + a : 0 \leq j < N\}$, given in Pochhammer / generalized-Stirling form. The single-valuedness of $S(n, c)$ is the character identity. No gaps remain.